Lecture 1: Principles of Economics (Chapter 1)

Applied Microeconomics — Master Notes · by Jakob V. Stangel

Required reading: Friberg, ch. 1 (*Introduction*).

Theme of the lecture (Butera): the "**economic way of thinking**" — economics studies how people **allocate scarce resources** among competing needs, using **three principles** (optimization · equilibrium · empiricism) and **models** (simplified, testable) to understand and predict behaviour.

> ★ **The heart of the chapter:** economics = **doing the best you can with scarce resources.** The toolkit is **three principles — Optimization · Equilibrium · Empiricism**; the two workhorses of optimization are **opportunity cost** and **marginal thinking**; and the method is building **positive (testable) models**, not making normative judgments. "*All models are useful, but all models are wrong.*"

The red thread. Everything starts from **scarcity** — without it we wouldn't need economics. From scarcity flows: you must **choose** (optimize) → every choice has an **opportunity cost** and is decided **at the margin** → many people choosing interact until no one wants to move (**equilibrium**) → and we test our claims about all this with **models + data** (empiricism). For any decision ask: *what's scarce · what's the best alternative given up (opportunity cost) · what does one more unit give me (marginal) · is this claim positive (testable) or normative (a value judgment)?*

Key concepts / "modes" to use

- **Scarcity** — limited resources vs unlimited wants; the reason economics exists.
- **The three principles: Optimization · Equilibrium · Empiricism.**
- **Opportunity cost** — the value of the *best alternative* you give up.
- **Marginal thinking** — decide by the value of *one more* unit; **law of diminishing marginal returns**.
- **Model** — a simplified representation of relationships between variables ("all models are wrong, but useful").
- **Positive vs normative** — a *testable* cause-effect claim vs a *value judgment*.

Section-by-section main points

What is economics?

Core: economics = **the study of how people allocate *scarce* resources over competing needs** — "people (try to) do the best they can for themselves with what they have."

- **Scarcity of two things:** the **things we want** (goods, services, jobs, education, relationships, recognition…) *and* the **means to get them** (money, **time, attention**, information, patience, willpower). Air to breathe isn't scarce → not an economic question.
- Every decision is shaped by **scarcity** (can't have it all), **other people** (you're not an island), and **the rules of the game** (you can't do whatever you want).

- **How societies allocate: central planners** (someone at the top decides who gets what — parents, government) vs **markets** (value set by supply & demand — from job markets to dating markets). Economics gives tools to predict how we decide and when markets do (or don't) allocate *efficiently*.

The three principles of economics

Core: the whole course rests on three ideas.

Principle	Meaning
① **Optimization**	We choose the **best feasible option** given our limited resources (money, time, attention, willpower are all scarce).
② **Equilibrium**	A situation in which **no one would benefit** from changing their *own* behaviour, given the choices of everyone else.
③ **Empiricism**	We build **models** → models generate **predictions** → we use **data/experiments** to test (falsify) them.

Optimization — opportunity cost + marginal thinking

Core: to optimise, use two tools — **opportunity cost** (weigh each use against its best alternative — "apples to apples") and **marginal thinking** (what does *one more* unit give me?).

Opportunity cost = the value of the best alternative you give up — nothing "free" is free if it uses a scarce resource.

OPPORTUNITY COST

$$\mathrm{OC}(\text{action}) = \text{benefit}(\text{action}) - \text{value of the best alternative given up}$$

- *Social media:* 1 hr/day on TikTok isn't free — you give up sport, study, work. Monetise the time with the hourly wage (avg **118 DKK** in Denmark):

YEARLY TIME-COST OF 1 HR/DAY ON TIKTOK

$$118 \text{ DKK/hr} \times 1 \text{ hr/day} \times 365 \text{ days} = \mathbf{43{,}070 \text{ DKK} / \text{year}}$$

- *Trip to Berlin (worked example):* drive = 1,280 DKK & 14 h; fly = 1,930 DKK & 3 h. Flying frees **11 h** you could work:

OPPORTUNITY COST OF DRIVING INSTEAD OF FLYING

$$\mathrm{OC}(\text{drive}) = (1{,}930 - 1{,}280) - (118 \times 11) = 650 - 1{,}298 = \mathbf{-648 \text{ DKK}} \implies \mathbf{fly}$$

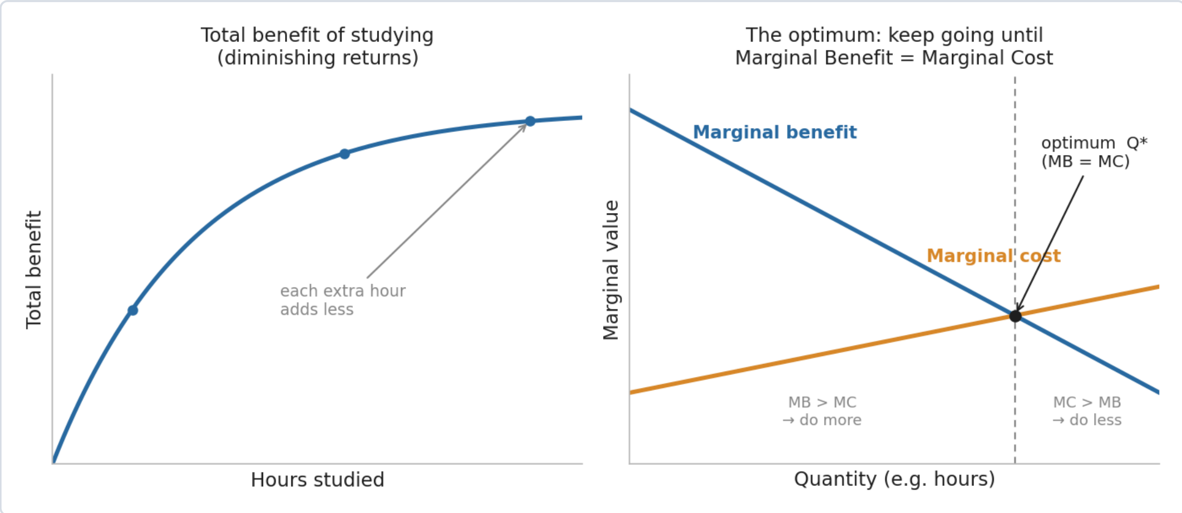
(Still a model — what if you enjoy the drive?)

Marginal thinking: decide by the value of the **next** unit — keep going while marginal benefit exceeds marginal cost; stop where they meet. That point is the **optimum**:

OPTIMUM CONDITION

choose the quantity where **Marginal Benefit = Marginal Cost** (MB = MC)

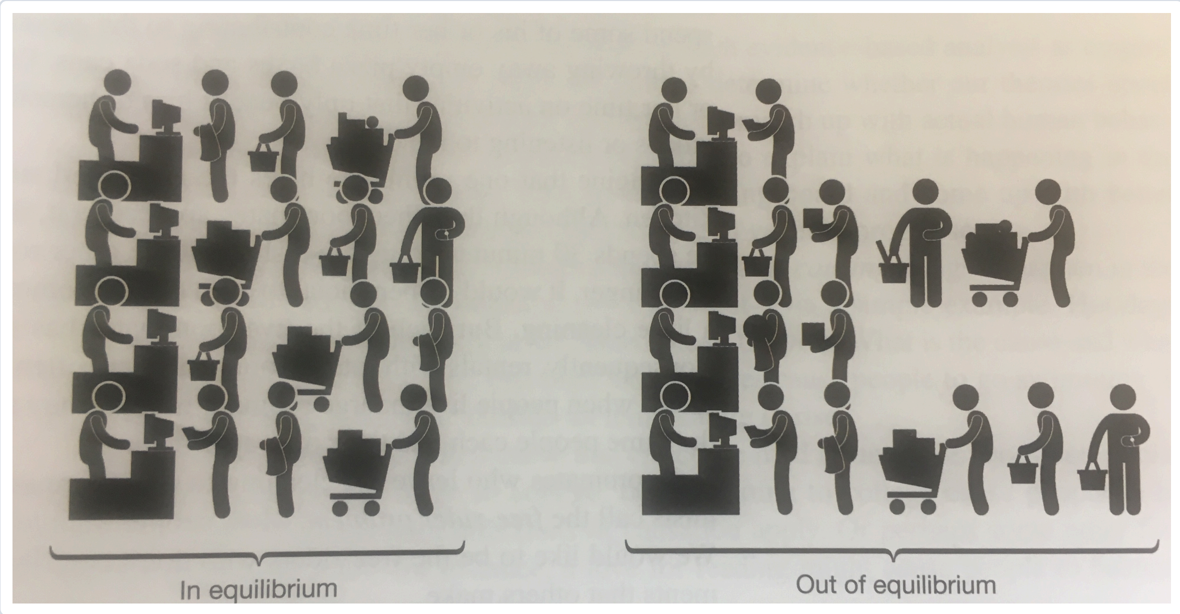
- *Night before the exam:* total benefit of studying rises then flattens; the **marginal benefit of each extra hour falls** (**law of diminishing marginal returns**). Optimal hours = where MB of one more hour just equals its MC.



How to read it (illustrative diagram — the graph the book draws in a later chapter). *Left:* **total benefit** rises but each extra hour adds less — the curve **flattens** (diminishing returns). *Right:* that flattening means **marginal benefit slopes down**; **marginal cost** slopes up. Left of the crossing, **MB > MC** so one more unit is worth it (*do more*); right of it, **MC > MB** (*do less*). The best you can do is the quantity where they meet — **the optimum Q*, where MB = MC.**

Equilibrium

Core: an **equilibrium** is a resting point — given what everyone else is doing, **no individual can make themselves better off by changing their own behaviour.** Market prices are the classic example (later lectures): they adjust until quantity supplied = quantity demanded and no one wants to move.

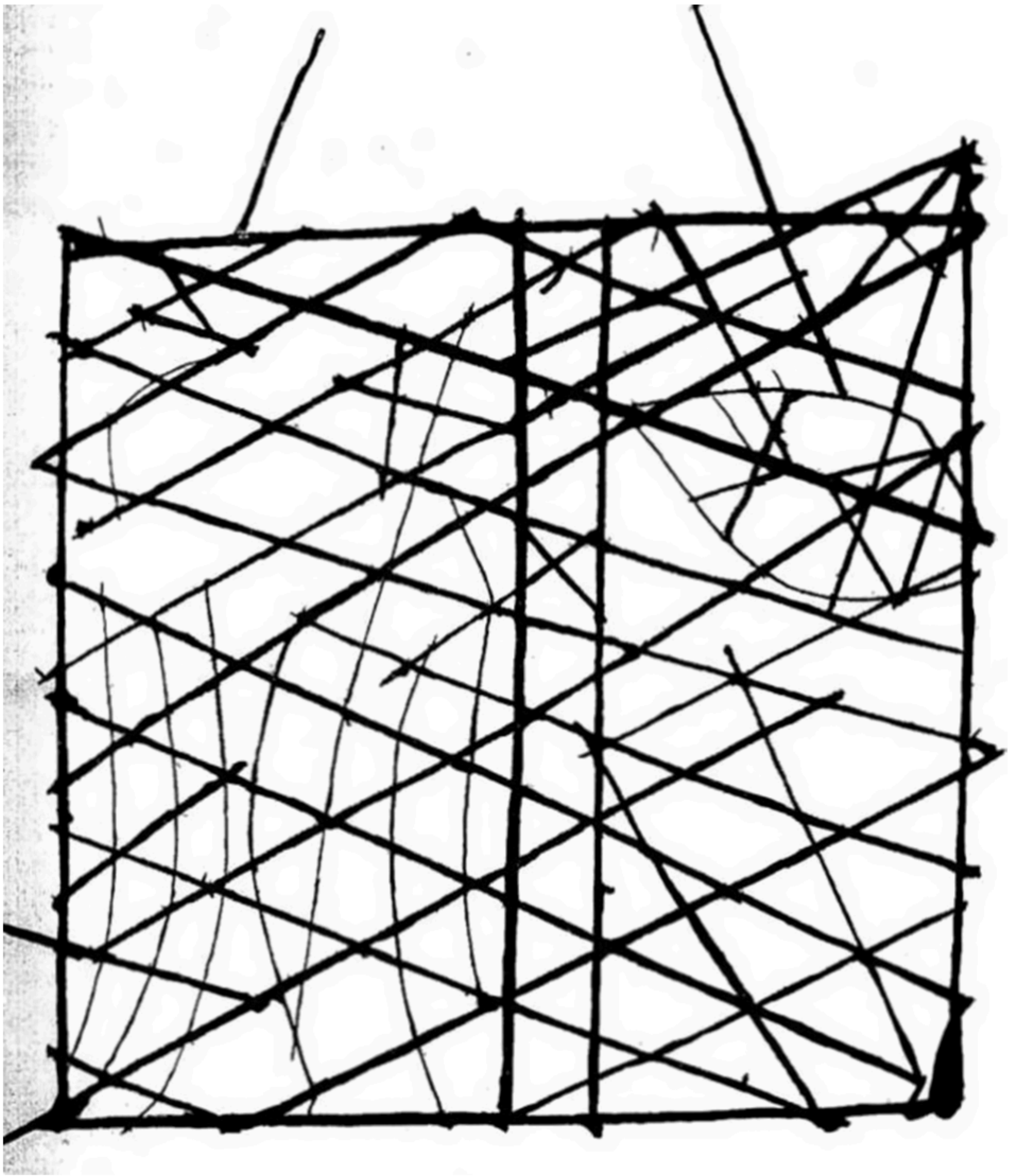


How to read it (equilibrium = supermarket queues, from the lecture). Each lane is a checkout; the queue length is how many people are waiting. **Out of equilibrium** (right) the lanes are uneven, so anyone in a long line can still gain by switching to a shorter one — people keep moving. **In equilibrium** (left) all lanes are equal, so **no one can do better by changing lane on their own** — that "no incentive to move" resting point is exactly what equilibrium means (and why market prices settle where supply meets demand).

Empiricism — models, and positive vs normative

Core: economics reasons with **models** and tests them against **data** — and it deals in **positive** (testable) statements, not **normative** (value) ones.

- **A model** expresses relationships between variables and makes **simplifying assumptions** (abstracts from reality). "*All models are useful, but all models are wrong*" — the art is choosing the right model (Keynes).



How to read it (Friberg Fig. 1.1 — what a "model" is). This is a Polynesian **stick chart**: sticks map the ocean's swell patterns, shells mark islands. It's not a literal picture of the Pacific — it **strips reality down to just the features you need to navigate.** That's the book's metaphor for an economic model: deliberately simplified, "wrong" as a full description, yet **good enough to steer decisions by.**

- **Positive vs normative:**
  - **Positive** = a **testable/falsifiable** cause-effect claim — "*each additional Starbucks in a ZIP code is associated with a 0.5% rise in housing prices*" (Glaeser et al. 2018).
  - **Normative** = a **value judgment** — "*gentrification is bad because it forces the poor out.*"
  - **Economics is about positive statements**, not normative ones.
- **Correlation ≠ causation:** in summer people eat more ice-cream *and* drown more — ice-cream doesn't cause drowning (a hidden cause: heat). A theory is only useful if it's **empirically falsifiable**.

- **Expanding models to fit data:** "people only give to charity because they care about others' well-being" fails the data (public spending doesn't crowd out giving 1-for-1 — Andreoni 1990; people give to *inefficient* charities). So we add "**joy of giving**" (utility from the act of giving itself).

Friberg's framing — allocation mechanisms (Ch. 1's example)

Core: a scarce good (e.g. **university places**) can be allocated many ways — and each has different effects on **efficiency** and **equity**. This is the book's opening illustration of "the role of economics."

- **Non-price mechanisms:** aptitude tests/interviews · secondary-school grades · open entry + hard exams · **waiting lists** · **queues** ("camp on the sidewalk") · **lotteries** · social class/legacy.
- **Price mechanisms:** everyone pays the **same fee** (set high enough to clear demand) · **auction** places to the highest bidders.
- Real systems **combine** several (fees + tests + interviews). Micro gives the tools to analyse the **trade-offs** (who gets in, how tied to parents' income, effect on effort).

Cases & examples

Each: *the example → the concept it teaches.*

- **1 hr/day of TikTok** → **opportunity cost** (≈43,070 DKK/yr at the average wage) — nothing using scarce time is free.
- **Drive vs fly to Berlin** → opportunity cost done properly: include the *value of time*, and driving turns out to *cost* 648 DKK more.
- **Studying the night before an exam** → **marginal thinking** + **diminishing returns**; optimise where MB = MC.
- **Ice-cream & drowning** → **correlation ≠ causation.**
- **Starbucks & house prices** (Glaeser 2018) → a **positive** statement; **gentrification** "**is bad**" → a **normative** one.
- **Charitable giving** (Andreoni 1990) → models must be **expanded to fit data** ("joy of giving").
- **University admissions** (Friberg) → **allocation mechanisms** (price vs non-price) and their equity/efficiency trade-offs.

Key terms

Term	Meaning
Scarcity	Limited resources vs unlimited wants — the basis of economics
Optimization	Choosing the best feasible option given constraints
Equilibrium	No one gains from changing their own behaviour, given others'
Empiricism	Build models → predict → test with data
Opportunity cost	Value of the best alternative given up
Marginal thinking	Deciding by the value of one more unit
Law of diminishing marginal returns	Each extra unit adds less than the last
Model	Simplified representation of relationships between variables
Positive statement	A testable/falsifiable cause-effect claim
Normative statement	A value judgment (good/bad, should/ought)
Allocation mechanism	Rule that decides who gets a scarce good (price/lottery/queue…)

Exam pointers

- State the **three principles** (Optimization · Equilibrium · Empiricism) and define each.
- **Compute an opportunity cost** properly (include the value of time) — the Berlin example is the model answer.
- Use **marginal thinking + MB = MC** to find an optimum, and name the **law of diminishing marginal returns**.
- Classify a statement as **positive vs normative**, and remember **economics = positive**; watch for **correlation ≠ causation**.
- Explain why economists use **models** ("all models are wrong, but useful") and give an **allocation-mechanism** trade-off (Friberg's university example).

★ Fun facts & memorable details

> Sticky lines from Lecture 1.

- Keynes: "*Economics is a science of thinking in terms of models joined to the art of choosing models which are relevant to the contemporary world.*"
- The course mantra: "***All models are useful, but all models are wrong.***"
- **1 hour of TikTok a day ≈ 43,070 DKK a year** in forgone wages (at Denmark's ~118 DKK average hourly wage).

- Driving to Berlin *looks* cheaper (1,280 vs 1,930 DKK) but, once you price the 11 lost hours, it actually **costs 648 DKK more** than flying.
- "*During the summer people eat more ice-cream and also drown more*" — the classic **correlation-≠-causation** trap (the hidden cause is heat).
- "**Joy of giving**" (Andreoni 1990): we don't donate only for others' welfare — we get utility from the *act* of giving, which is why public spending doesn't crowd out private giving 1-for-1.
- Learning micro is "*a first course in a new language*" — a bit hard, not immediately intuitive, needs practice.